

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**Seventh Semester of B. Tech. (CL) Examination****March - April, 2018****CL408.01 Irrigation & Hydraulics structures****Date: 04.04.2018, Wednesday****Time: 10:00 a.m. To 1:00 p.m.****Maximum Marks: 70****Instructions:**

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures with pencil only wherever required.
4. Use of scientific calculator is allowed.

SECTION – I**Q-1** Select the correct answer for the following and rewrite the complete correct statement**[07]****1** The duty is largest

- | | |
|---------------------------------|------------------------|
| (a) at the head of water course | (b) on the field |
| (c) at the head of a main canal | (d) same at all places |

2 The area on which crops can be grown satisfactorily is known as

- | | |
|------------------------|-----------------------|
| (a) Gross command area | (b) Duty area |
| (c) Farm region | (d) none of the above |

3 The crop evapotranspiration when the soil moisture content is upto field capacity is called

- | | |
|-----------------------------------|-----------------------------------|
| (a) potential evapotranspiration | (b) equivalent evapotranspiration |
| (c) pellicular evapotranspiration | (d) none of the above |

4 The most suitable material for central impervious core zone of an embankment dam is

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|----------|-----------------|----------|-------------------------------|
| (a) clay | (b) coarse sand | (c) silt | (d) clay mixed with fine sand |
|----------|-----------------|----------|-------------------------------|

5 As compared to earthen dam, concrete gravity dams

- | | |
|-------------------------------------|------------------------------------|
| (a) are costlier | (b) requires sound rock foundation |
| (c) are less susceptible to failure | (d) all of the above |

6 The major resisting force in gravity dam is

- | | |
|--------------------|---------------------|
| (a) water pressure | (b) wave pressure |
| (c) self weight | (d) uplift pressure |

7 The uplift pressure can be controlled by constructing

- | | |
|---|----------------------|
| (a) drainage channel between dam and its foundation | (b) drainage gallery |
| (c) cutoff at upstream face | (d) all of the above |

Q-2(a) During a particular stage of the growth of a crop, consumptive use of water is 2.8 mm/day. **[04]**
Determine the interval in days between irrigations, and depth of water to be applied when the amount of water available in the soil is: (i) 25%, (ii) 50% (iii) 75%, and (iv) 0% of the maximum depth of available water in the root zone which is 80 mm. Assume irrigation efficiency to be 65%

(b) Attempt any two **[10]**

- 1 Define 'Delta' and 'Duty'. Derive the mathematical relationship between them.
- 2 Briefly explain Swedish circle method of slope stability.
- 3 What do you understand by 'gravity dam'? How elementary profile of a low gravity dam is determined ?
- 4 Write a brief note on : Galleries in Gravity dam.

Q-3 (a) A trapezoidal gravity dam is 6 m high, 1.5 m wide at top and 4.5 m wide at bottom with vertical water face. Determine the normal stress at toe and heel of the dam for reservoir empty and full condition. Take Weight density of concrete = $2.4 \times 10^4 \text{ N/m}^3$ and $c=1$. Consider unit length of dam. Neglect the earthquake effects. **[06]**

(b) Attempt any two **[08]**

- 1 Classify the method of irrigation and write a brief note on any one of them.
- 2 What is meant by consumptive use of water?
- 3 What is an earthen dam ? How it fails ? Explain any one cause of failure of an earth dam with neat sketches.

SECTION – II

Q.4 Match the following : **[07]**

SECTION A

- 1 Exit gradient for a weir without downstream sheet pile is
- 2 Continuous seepage under a weir causes
- 3 Relationship between elevation and reservoir storage is
- 4 Canal falls helps for
- 5 For An aquaduct the canal flows
- 6 In branching type canal system discharge in direction of flow
- 7 For a canal syphon, canal flows

SECTION B

- (a) decreases
- (b) over the drain
- (c) Energy decapitation
- (d) below the drain
- (e) piping action
- (f) infinity
- (g) increasing and nonlinear

Q.5

- (a) Explain the terms (i) inverted filter (b) launching apron with neat sketches. [04]
- (b) Attempt Any Two: [10]
- 1 Explain the important factors to be considered while selecting a reservoir site.
 - 2 Differentiate between weir, and barrage. How they are different from a dam ?
 - 3 Design an irrigation canal in alluvial soil carrying a discharge of 35 cumecs. Take Lacey's silt factor as one. Assume trapezoidal section with side slope 0.5 H to 1 V. Use Lacey's regime theory.

Q.6

- (a) Draw a typical mass curve and demand curve applied for determining reservoir capacity. [02]
- (b) What is cross regulator and head regulator? Why they their functions in a canal system [03]

OR

- (b) With the help of a neat sketch explain the various components of a Sarda type canal fall. [03]
- (c) Attempt Any Three: [09]
- 1 Explain reservoir sedimentation and methods for reducing sedimentation.
 - 2 Explain the importance of canal lining also list the various types of canal lining.
 - 3 Explain the channel design based on Kennedy's silt theory.
 - 4 Briefly explain the function of a canal outlet. List the various types of outlets.
